MOODY'S

Using the GCorr™ Macro Model for Multi-period Stress Testing of Credit Portfolios

Abstract

Moody's Analytics Global Correlation Model (GCorr™) Macro model links borrower credit quality to macroeconomic variables, enabling multi-period probability of default (PD) and loss given default (LGD) projections under different economic scenarios. The model can be used in a variety of applications, including stress testing for Comprehensive Capital Analysis and Review (CCAR) and Dodd-Frank Act Stress Tests (DFAST), impairment calculations, reverse stress testing, and conditional simulation.

This document presents the results for the GCorr Macro 2023 model, the most recent model estimated using data up to 2023 Q1. As the overall economy stabilizes from the negative impact of the inflation, milder relationships between macro-variables and credit factors, especially in gross domestic product (GDP) and unemployment types of macroeconomic variables, have been observed. As a result, the overall correlation levels are slightly lower than GCorr 2022's and have receded close to pre-pandemic levels.

This document also describes the different validation exercises performed. The main focus is to present several validation studies for portfolios from various regions of the world and various asset classes. In particular, we compare how GCorr Macro projected losses under historical scenarios compare to different benchmarks. We also present the losses projected by the GCorr Macro for C&I and CRE portfolios under the Federal Reserve Board´s CCAR scenarios.

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Contents

1. Overview 3
2. Correlations of Macroeconomic Variables with GCorr Factors 4
   2.1. Correlation Summary Statistics 4
   2.2. Cross-sectional Correlation Variation Across Industries 5
   2.3. Cross-sectional Correlation Variation Across Countries 5
3. Validation of the GCorr Macro Model with Historical Scenarios 7
   3.1. U.S. Large Corporate and U.S. SME Portfolios 7
   3.2. International Corporate Portfolios 15
   3.3. U.S. Commercial Real Estate Portfolios 19
4. Projected CCAR Losses Based on the GCorr Macro Model 22
   4.1. U.S. Large Corporate and SME Portfolios 22
   4.2. U.S. Commercial Real Estate Portfolios 25
5. Summary 26
A. Variable Selection Results for Corporate Exposures 27
B. List of Macroeconomic Variables for Sovereign Countries 29

1. Overview

The GCorr Macro model is a correlation-based model designed by Moody's Analytics to project scenario-based probability of default (PD) and loss given default (LGD). The model has a number of applications, including stress testing for Comprehensive Capital Analysis and Review (CCAR) and Dodd-Frank Act Stress Tests (DFAST), impairment calculations, reverse stress testing, and conditional simulation. The GCorr Macro model is an extension of Moody's Analytics GCorr model (Global Correlation model), which describes the correlations across credit quality changes of counterparties in a credit portfolio. The GCorr Macro model links systematic credit risk factors from the GCorr model to observable macroeconomic variables. In this paper, we discuss how to use the GCorr Macro model for multi-period stress testing of credit portfolios.

When using the GCorr Macro model, it is important to properly parameterize all of the inputs of the model: PD, LGD, correlation parameters, and scenarios. Moody's offers off-the-shelf parameterization for unconditional PD and LGD term structures. This parameterization is provided through the Rating to PD Term Structure model and the LossCalc model, respectively.[1] Moody's Economics.com provides the forecasts for various macroeconomic scenarios.

The remainder of this paper provides guidance on the following topics related to using the GCorr Macro 2023 model:

- Section 2 summarizes the GCorr Macro 2023 parameters, including correlations between GCorr Corporate factors and macro-economic variables. It also illustrates how we use these parameters to conduct stress testing.
- Section 3 presents validation exercises of the GCorr Macro 2023 model with commercial and industrial (C&I) and Commercial Real Estate (CRE) portfolios under various historical scenarios.
- Section 4 shows the losses projected by the GCorr Macro 2023 model for the C&I and CRE portfolios under the Federal Reserve Board's CCAR scenarios.
- Section 5 summarizes the document.

[1]For a discussion on the underlying data and validation around the Rating to PD Term Structure model and the LossCalc model, see ("RiskCalc Private Firm Converter" 2018) and Zhuang and Dwyer (2016).

2. Correlations of Macroeconomic Variables with GCorr Factors

The GCorr Macro framework models the correlations between GCorr credit risk factors and macroeconomic variables. This section provides an overview of the estimated GCorr Macro 2023 parameters. The GCorr factors are drivers of credit risk and these drivers vary by asset class. For example, the systematic portion of corporate credit risk is impacted by the counterparty's country and industry risks. Commercial real estate credit risk can be decomposed by geography and property type, and retail credit risk by geography and product type. Each asset class has its corresponding set of systematic factors that are subsequently linked to macrovariables.

2.1 CORRELATION SUMMARY STATISTICS

Table 1 presents summary statistics of correlations between the 61 GCorr composite factors representing U.S. industries with several U.S. macrovariables.

Table 1 Summary Statistics of Correlations between Select U.S. Macroeconomic Variables and 61 GCorr Composite Factors Representing U.S. Industries

	CORRELATION WITH THE 61 U.S. GCORR CUSTOM INDEXES			
MACROECONOMIC VARIABLE	AVERAGE	STD.DEV.	5TH PERCENTILES	95TH PERCENTILES
Economic activity				
Real GDP	35%	2%	31%	37%
Unemployment rate	-32%	2%	-34%	-28%
Financial markets				
BBB Spread	-52%	3%	-56%	-46%
Dow Jones Total Stock Market Index	54%	3%	48%	58%
VIX - Stock Market Volatility	-35%	2%	-37%	-31%
Real estate markets				
House price index	21%	1%	19%	23%
CRE price index	23%	1%	21%	25%
Commodity				
Oil price	18%	8%	10%	30%
Consumer prices				
Consumer price index	14%	1%	12%	15%
Disposable income				
Real personal disposable income	0%	3%	-5%	5%
Nominal personal disposable income	-1%	3%	-7%	5%

The variables in Table 1 are divided into several categories according to their definition and interpretation. As the table indicates, some variable types have a stronger relationship to the credit risk factors than others. Namely, some economic activity variables (GDP, unemployment rate, and so on) and some financial market variables (stock market index, VIX, corporate spread), exhibit the highest correlation with the factors. Both the magnitude and signs of the correlations are consistent with economic intuition. As the overall economy continues to stabilize from the impact of the recent unexpected events (Pandemic, geopolitical tensions, inflation, and so on), milder relationships between macro-variables and credit factors, especially in gross domestic product (GDP) and unemployment types of macroeconomic variables, has been observed compared to 2022 model.

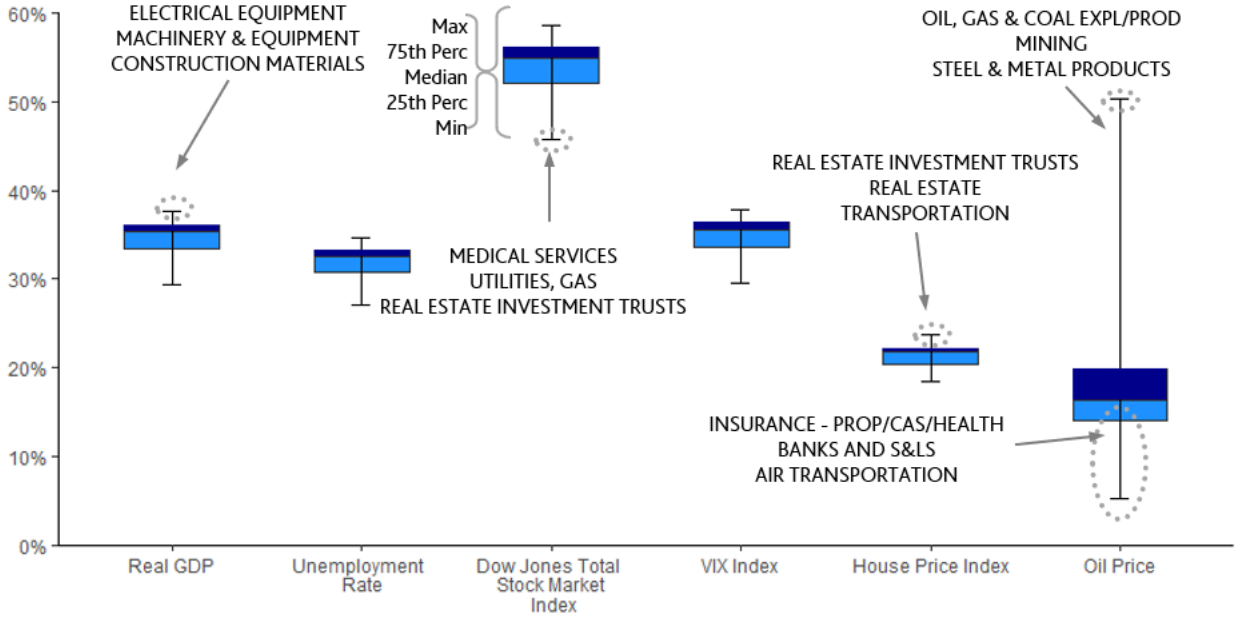
Other variables, for example, real and nominal personal disposable income, have very low correlations with the

credit risk factors. One reason for this is that disposable income incorporates effects of government policies, such as tax rebates included in past government stimulus packages. Stimulus packages were approved in response to worsening economic conditions in 2001, 2008, 2009, and 2020. As a result, disposable income might increase during quarters when the economy deteriorates and credit risk factors experience negative shocks. This leads to the low single-digit correlations.

2.2 CROSS-SECTIONAL CORRELATION VARIATION ACROSS INDUSTRIES

Figure 1 illustrates the variation in correlations for several macroeconomic variables across several major industries. For example, we observe a high variation for Oil Price. The Oil, Gas & Coal Expl/Prod and Mining industries, with revenues linked to oil and commodity prices, show the highest correlations with Oil Price. At the other end of the spectrum, we see the Air Transportation industry, for whom crude oil products are a cost of flying, has a low correlation with Oil Price. When Oil Price decreases, airlines benefit because it becomes less expensive for them to operate.

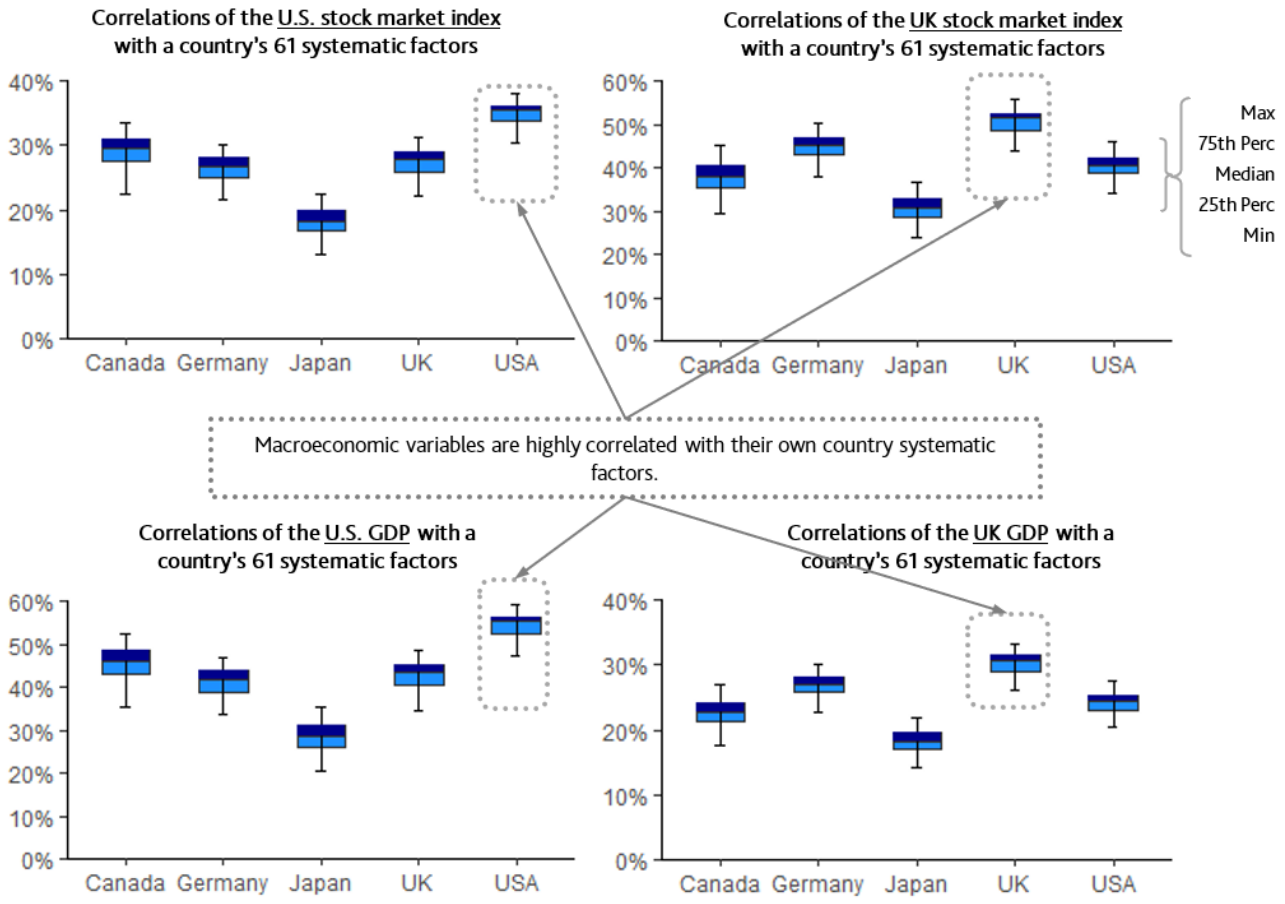
Figure 1 Cross-sectional variation in correlations of select U.S. macroeconomic variables and GCorr systematic factors representing 61 U.S. industries (correlations of Unemployment Rate and VIX are scaled by -1).



2.3 CROSS-SECTIONAL CORRELATION VARIATION ACROSS COUNTRIES

Figure 2 illustrates the correlations of two U.S. and UK macroeconomic variables with the systematic factors of several countries. The U.S. macroeconomic variables tend to be more closely correlated with U.S. credit risk factors than with other countries' factors, which is in line with economic intuition. Moreover, the U.S. macroeconomic variables have a larger impact on, for example, Canadian factors than Japanese or German factors. The UK macroeconomic variables also have high correlations with the UK factors relative to the other countries' factors. Although the figure shows macroeconomic variables for two countries only, we can also draw similar conclusions for other countries.

Figure 2 Cross-sectional variation in correlations of two U.S. and UK macroeconomic variables with GCorr systematic factors representing 61 industries in several countries.



3. Validation of the GCorr Macro Model with Historical Scenarios

This section presents several analyses illustrating levels and patterns in credit portfolio losses produced by the GCorr Macro model over recent economic episodes. Our objective is to assess how the scenario-based probability of default (PD) compares to our benchmarks. In particular, this section assesses how scenario-based PD and losses compare to various benchmarks to understand how different aspects of the modeling framework impact the losses. This type of analysis contributes to the process of GCorr Macro model validation.

For this analysis, we must first select the appropriate set of macroeconomic variables for a given sample portfolio. We refer to this set of variables as "model". The selected model and its performance for historical scenarios are portfolio-specific, so the analysis is carried out for sample portfolios across various regions and asset classes.

In this section, we present and discuss the backtesting results produced by the GCorr Macro model for the following:

- Section 3.1: U.S. large corporate and U.S. small-to-medium enterprise (SME) portfolios
- Section 3.2: International corporate portfolios
- Section 3.3: U.S. commercial real estate portfolio

3.1 U.S. LARGE CORPORATE AND U.S. SME PORTFOLIOS

We use two stylized credit portfolios to validate the GCorr Macro model for U.S. Corporates: a portfolio of exposures to U.S. large-listed corporates (U.S. Large Corporates portfolio) and a portfolio of exposures to U.S. Small- and Medium-sized Enterprises (U.S. SMEs portfolio). Table 2 summarizes portfolio characteristics.

Table 2 Stylized Portfolios Used for Validation

PORTFOLIO	U.S. SME PORTFOLIO	U.S. LARGE CORPORATES PORTFOLIO
Types of Counterparties	U.S. Small- and Medium-sized Enterprises (non-financial)	U.S. large-listed corporates (firms constituting 99% of total liabilities issued by listed firms)
Exposure Pooling	3111 pools of loans Loans are pooled by 61 SME industries and 51 states	61 pools of loans Loans are pooled by 61 GCorr industries
R-squared	Weighted average R-squared = 7.17% [a]	Weighted average R-squared = 34.73% [b]
Probability of Default	Time varying PD[c]	Time varying PD[d]
Loss Given Default for Projections	LGD = 50% [e]	Two cases - LGD = 100% (used for PD benchmarking) - LGD = 40% (used for projections)

[a] Source: GCorr 2023 SME, R-squared values by industries and states.
[b] Source: GCorr 2023 Corporate, large firm average R-squared values by industries.
[c] Average pool-level RiskCalc U.S. EDF. The time-varying PD is used for backtesting.
[d] Average pool-level CreditEdge U.S. EDF. The time-varying PD is used for backtesting.
[e] The LGD of 50% matches the LGD from the paper 'Stress Testing Probability of Default for Private Firm C&I Portfolios: RiskCalc Plus Stress Testing PD and LGD Model (granular approach) - United States v4.0 Corporate Model' by Chen, et al., 2014, which we use as one of the benchmarks for the GCorr Macro model.

All calculations are based on the GCorr Macro 2023 model[2], GCorr 2023 Corporate model[3], and GCorr 2023 SME model[4]. In the analyses where we perform scenario-conditioning on LGD, the LGD variance is parameterized with $k = 4$. Furthermore, we set the PD-LGD correlation parameters to $RSQ_{RR} = 38\%$ and $\rho_{A,RR} = 35\%$. In the case of unstressed LGD analyses, we fix the LGD level either at 40% for U.S. large corporates or 50% for the U.S. SME portfolios.

Before we begin the validation exercises, we must select the macroeconomic model to use. To do so, we first conduct a univariate statistical analysis of the economically- relevant macro variables. We discard all macro variables with factor coefficients that are either insignificant or that have an unintuitive sign. The remaining macro variables are then in a multivariate analysis. Again, we remove models with unintuitive signs or insignificant coefficients (using 10% significance level) and rank the models that pass these tests according to their adjusted pseudo R-squared and backtesting performance. For the case of US Large portfolios we select the following macroeconomic variables: US Unemployment, US Equity, US BBB Spread[5]. For the case of US SME portfolios we select the following macroeconomic variables: US Unemployment, US Equity, US GDP.

The next two subsections present the backtesting results for US Large and SME portfolios. First, we describe the GCorr Macro model scenario-based losses based on historical economic episodes from 2001 through 2023. We want to find out how the losses vary across the economic episodes within this period. The second part of the validation analysis focuses on the levels of scenario-based losses projected by the GCorr Macro model and compares these losses to losses provided by various benchmarks. We report nine-quarter cumulative losses, which correspond to the time horizon considered by CCAR.

Time Series Patterns in Scenario-Conditioned Expected Losses

In the first set of validation exercises, we compare stressed expected losses produced by the GCorr Macro model across various economic episodes. In particular, we focus on the levels of losses during the recent financial crisis.

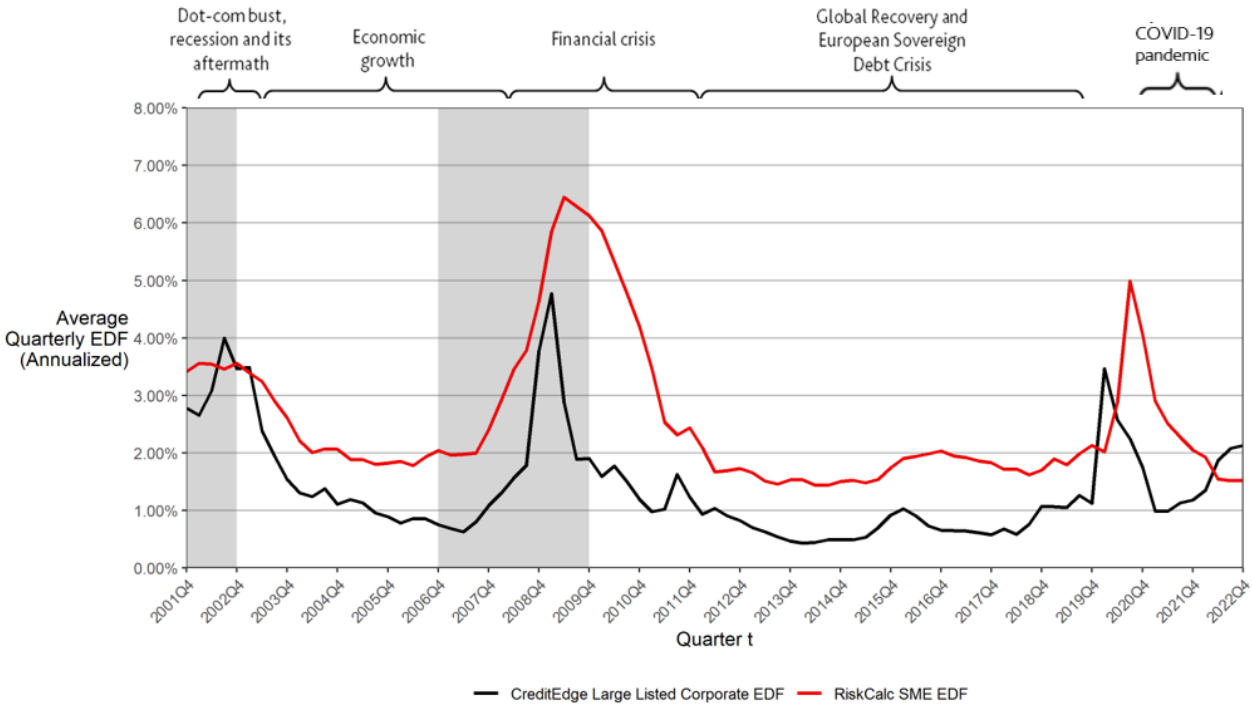
We consider the period 2001 through 2023, which includes five distinct episodes:

- Dot-com bust, the recession of 2001, and its aftermath
- Period of economic growth, approximately mid-2003 through mid-2007
- 2008-2009 financial crisis
- Global recovery and Eurozone sovereign debt crisis of 2010-2019
- COVID-19 pandemic, the Great Lockdown in 2020, and its aftermath

Figure 3 displays these economic episodes using the average CreditEdge™ benchmark expected default frequency (EDF) for U.S. large public firms and the average RiskCalc™ EDF value for the U.S. SME portfolios. EDF values are high during periods of economic distress, but they fall to low levels during periods of economic growth. Comparing both measures, we see that large corporate EDF measures are more pronounced during downturns but the SME EDF levels are generally higher than those of large corporates.

[2]For calculation details, please refer to Huang et al. (2017)
[3]For model details, please refer to Lkhagvajav, Xiao, and Lanfranconi (2024)
[4]For model details, please refer to Xiao, Pei, and Lanfranconi (2024)
[5]For more details about the variable selection process please refer to Huang et al. (2017).

Figure 3 Average quarterly CreditEdge benchmark EDF value for U.S. large public firms and average RiskCalc EDF value for U.S. SME firms.



Our first goal is to understand the sensitivity of GCorr Macro projected losses for U.S. SME and U.S. large corporate portfolios to changing macroeconomic conditions. In Figure 4 and Figure 5, we plot the nine-quarter cumulative stressed expected losses estimated with the GCorr Macro model starting from 2001, while using an unconditional PD term structure. Controlling for fluctuations in unconditional PD values allows us to isolate the effect on expected losses that are coming from the change in the historical macro-economic environment. The scenarios are defined using historical values of the macroeconomic variables from U.S. Unemployment, U.S. Equity, and U.S. BBB Spread for U.S large portfolio and U.S. Unemployment, U.S. Equity, and U.S. GDP for U.S SME portfolio. The portfolio characteristics remain the same across the period.

Figure 4 Cumulative nine-quarter expected losses, unconditional and scenario-based, for the U.S. SME portfolio.

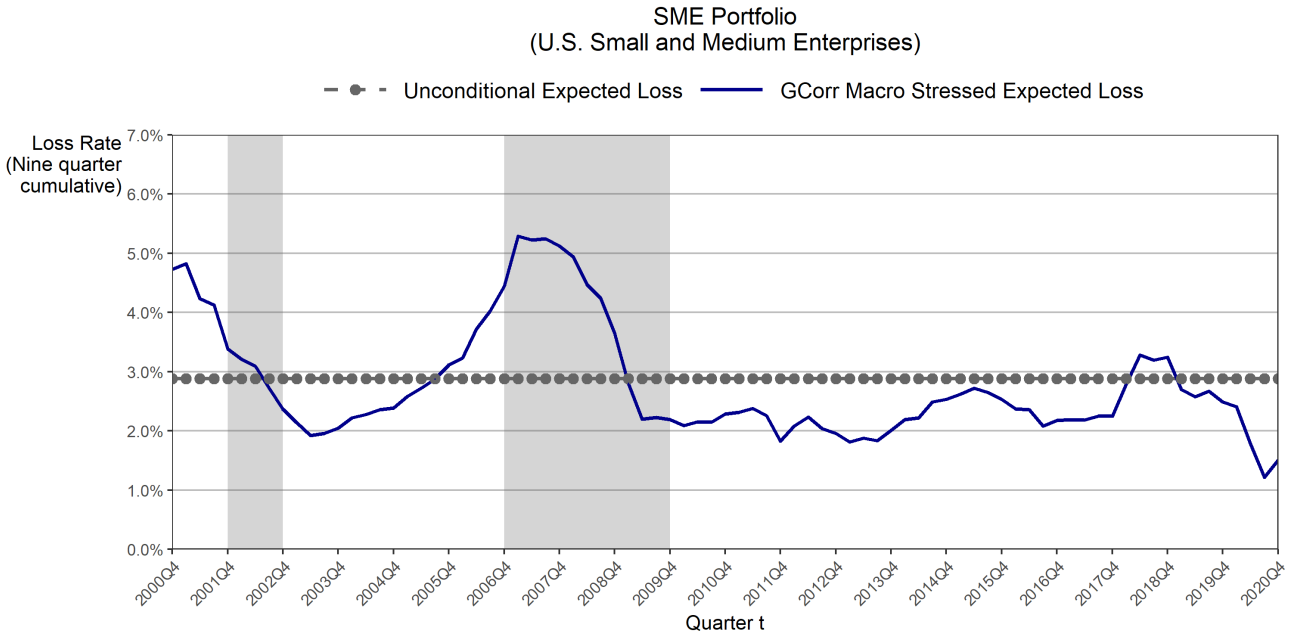


Figure 5 Cumulative nine-quarter expected losses, unconditional and scenario-based, for the U.S. Large Corporates portfolio.

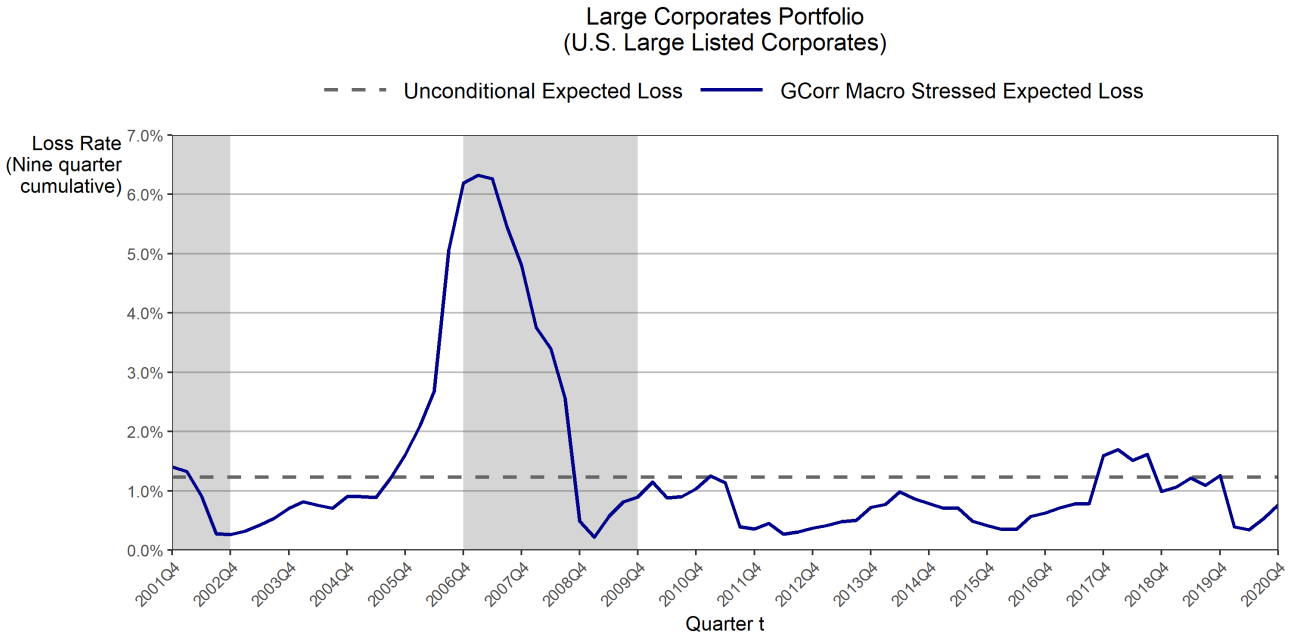


Figure 4 and Figure 5 show similar patterns in terms of time series dynamics:

- Losses are higher during periods of economic distress and lower during the period of economic growth.
- The GCorr Macro model produces high conditional credit parameters and expected losses for those nine-quarter periods when the scenario assumes most negative shocks to the macroeconomic variables. As a result, the series in Figure 4 and Figure 5 peak for the nine-quarter period 2007Q1 through 2009Q1.

» The scenario-conditioned expected losses are higher during the recent financial crisis than during the early 2000s recession. We attribute this result to macroeconomic variable dynamics during these two episodes. As an example, we show time series of log changes in the Unemployment Rate and the Dow Jones Total Stock Market Index in Figure 6. Both variables, but especially Unemployment Rate, experienced larger quarterly shocks during the global financial crisis. This also applies to U.S. BBB Spread.

» The change in Unemployment Rate during the first half of 2020 was unprecedented in its scale. The COVID-19 crisis also led to dramatic swings in the stock market index, as well as in the U.S. BBB Spread. These stressed macro barometers lead to higher scenario-conditioned expected losses for the nine-quarter period 2017Q4 through 2018Q4.

Figure 6 Quarterly log changes in Unemployment Rate, Dow Jones Total Stock Market Index, and BBB Spread.

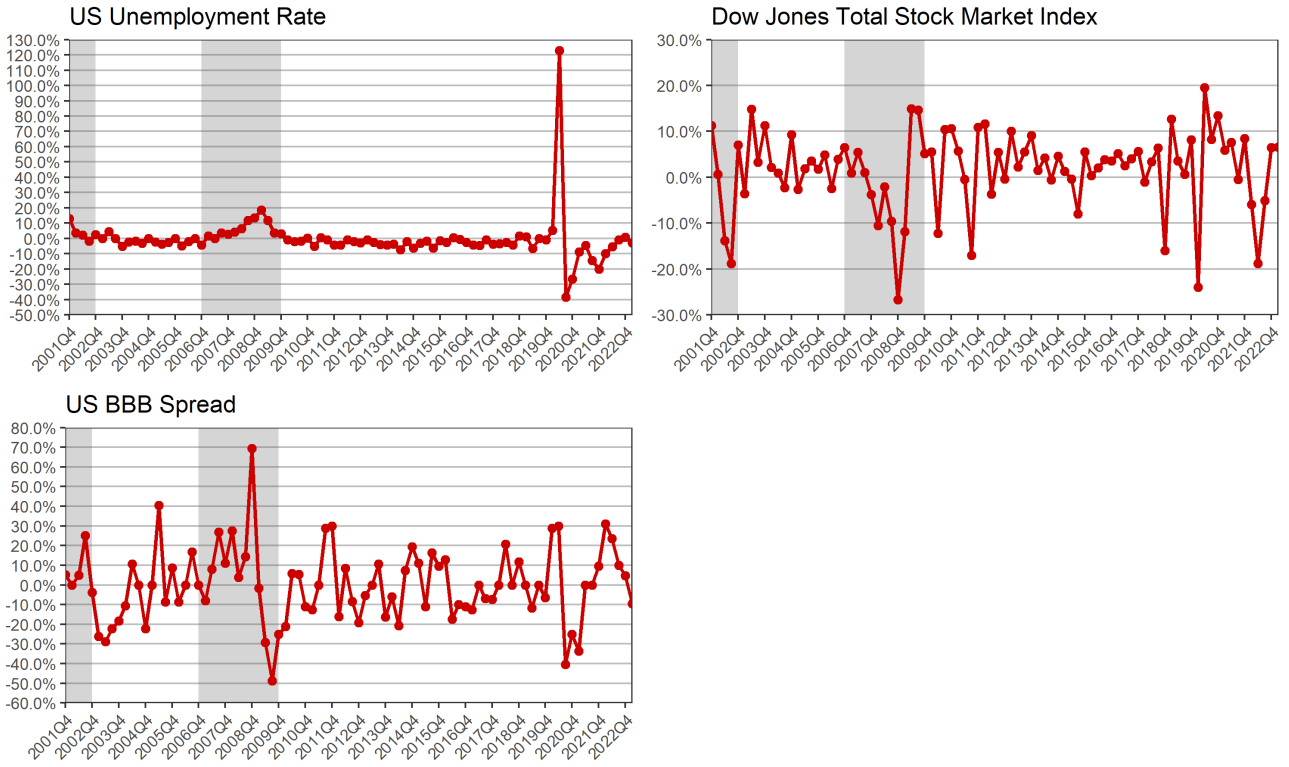


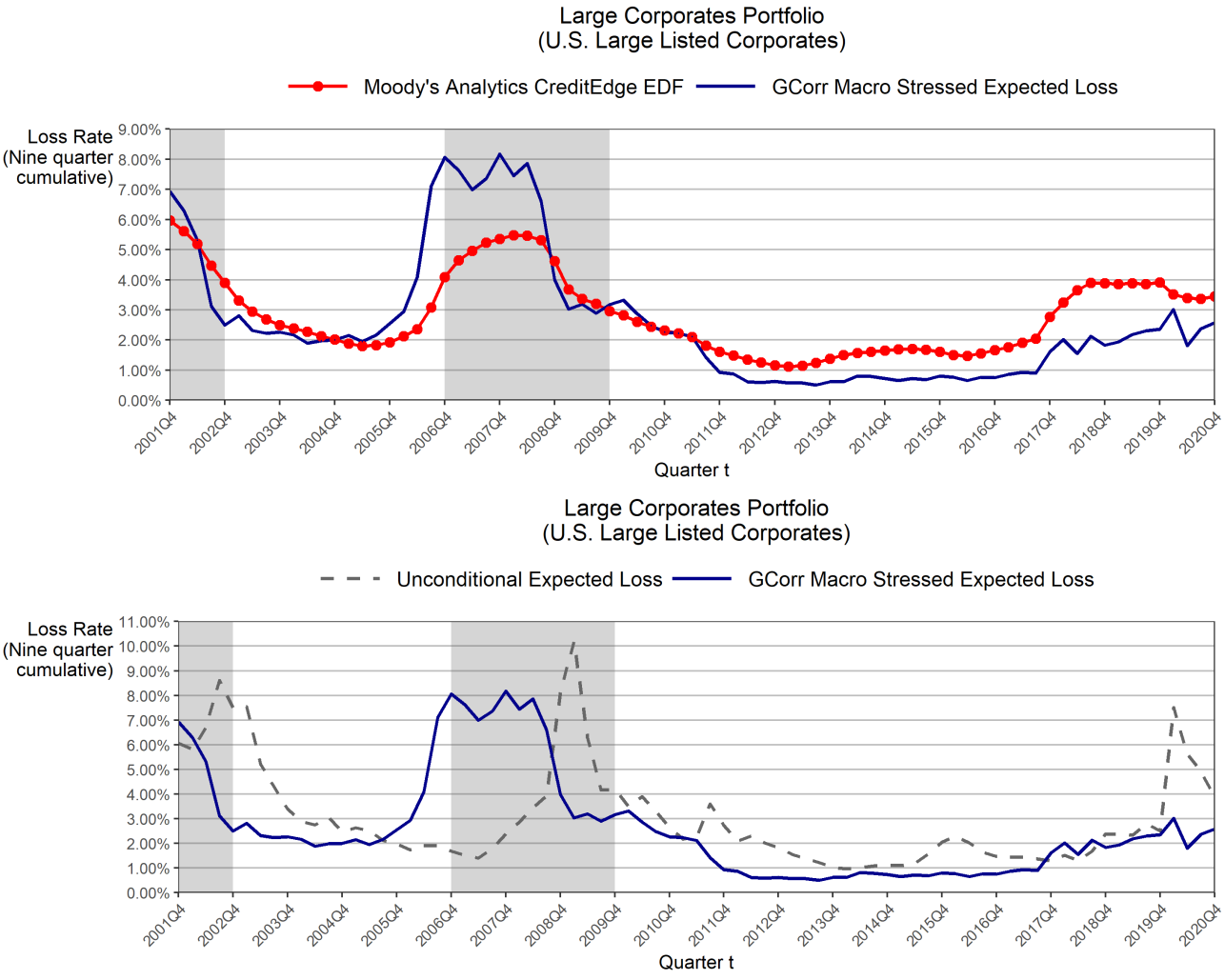
Figure 4 and Figure 5 also indicate that the time series of losses for the Large Corporate portfolio fluctuate more than for the SME portfolio. Specifically, the scenario-based losses on the large portfolio became about six times larger than the unconditional expected losses during the Global Financial Crisis. For the SME portfolio, they were close to two times larger. We can attribute this difference to the difference in R-squared values of the portfolios. The U.S. large corporates have substantially higher average R-squared values than the SMEs, 34.73% and 7.17%, respectively. These values imply that economic distress of a given magnitude will have a larger impact on U.S. large corporates. This is consistent with what we observed from default statistics across the globe: Commonly the unconditional losses for smaller firms are higher than the large corporates, although they are less sensitive to systematic failures such as the Global Financial Crisis.

Next, we evaluate historical performance of the GCorr Macro model and the selected models for U.S. large corporates and SMEs, conducting a backtesting exercise. In this exercise, we study how well the predicted model results match up with the historical behavior of certain benchmarks. The aim is to validate whether our model can explain observed historical movements in stressed expected losses and probability of default proxies in a plausible way.

Beginning with the U.S. Large Corporates portfolio, Figure 7 shows the backtesting results of the scenario-

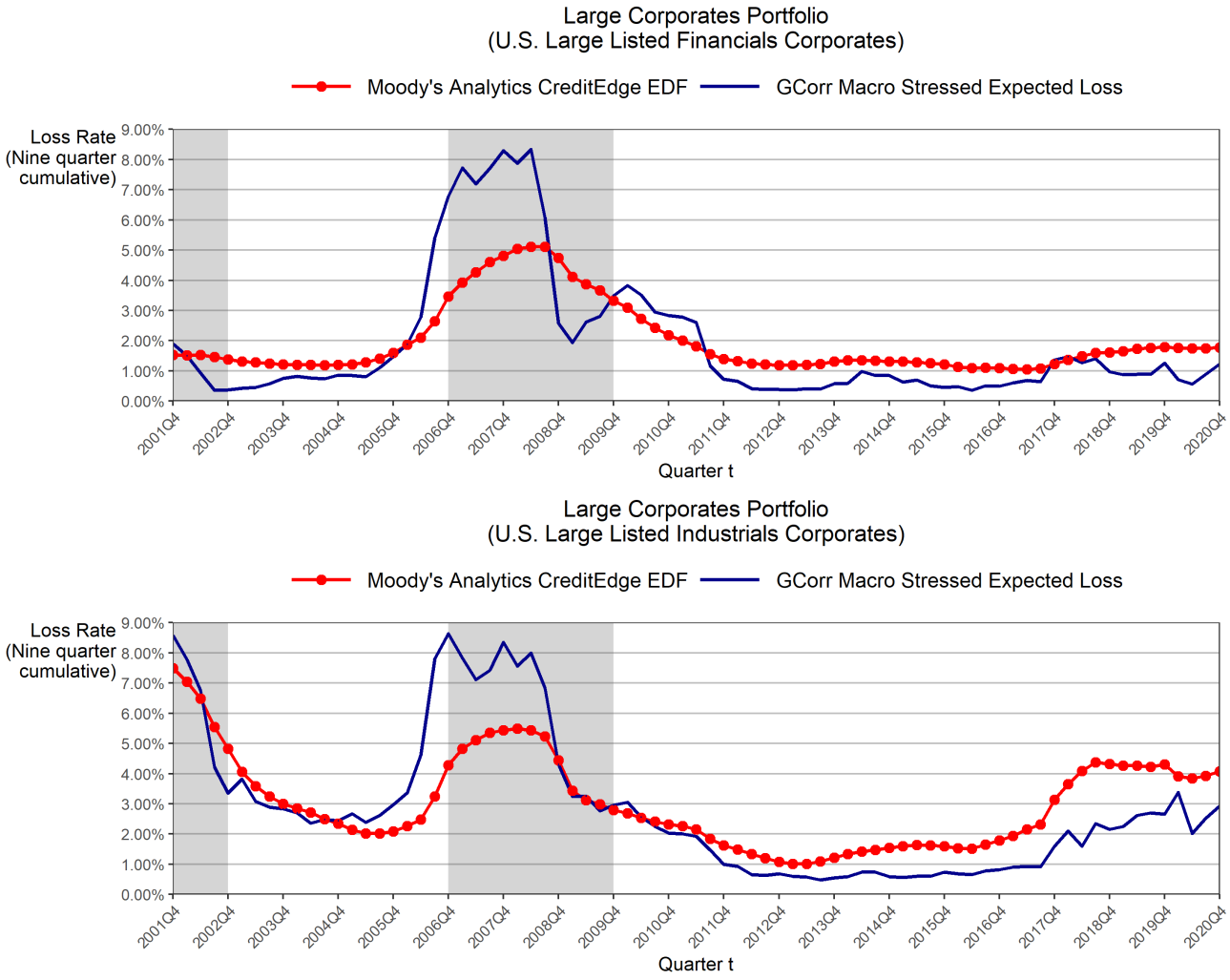
based probability of default. Here, we focus on the top model through variable selection and compare it to the historical movement of the nine-quarter average CreditEdge benchmark EDF values for large U.S. public firms. We obtain the GCorr Macro model scenario-based PD by computing the scenario-based loss with a constant LGD of 100% and a time-varying unconditional (input) PD for each of the portfolio's loan pools. We determine the unconditional PD for each nine-quarter period using EDF values as of the beginning of that period. In the upper plot, the Moody's CreditEdge™ EDF value is used as the benchmark for these scenario-based PD values. It is constructed by computing for each quarter the average EDF value for the sample's U.S. firms. Afterward, for each point in time, we cumulate losses over the next nine quarters. In the lower plot, the backtesting result is compared with unconditional expected loss, to provide another view from the angle of real-time historical EDF values.

Figure 7 Backtesting the GCorr Macro model scenario-based loss for a U.S. large corporate portfolio — time-varying input PD and LGD=100%.



Assessing Figure 7, we see that the overall time series patterns of the benchmark are matched by the top GCorr Macro model. Three periods of high PDs in stressed periods (2001, 2007 through 2008, and 2020) are observable for the GCorr Macro models, with the first arising from the dot-com bust, the second occurring during the financial crisis, and the third due to the COVID-19 pandemic. The unconditional loss shows a time lag in its spike compared to the stressed variables because, by design, it does not take into account the future macroeconomic environment of the following quarters. In recent periods, the EDF remains at a high level, in particular for industrial firms with supply chain challenges due to the aftermath of the pandemic.

Figure 8 Backtesting of the GCorr Macro model scenario-based loss — top U.S. model of large U.S. Financials and Industrials portfolio with time-varying PD term structure and LGD=100%.



In Figure 8, we split the U.S. Large Corporates portfolio into Financial and Industrial subportfolios. We observe several differences in the scenario-based PD behavior over time:

- First, we see that the dot-com bust represents the most significant PD increase for the industrial subportfolio, overtaking the increase seen during the financial crisis and the COVID-19 pandemic. For the Financials subportfolio, the dot-com bust leads to a moderate increase of roughly 1.5% in PD, and it is overshadowed by the huge and prolonged increase during the financial crisis.
- In the more recent economic crisis, both portfolios benefit from the swift reaction from the Federal Reserve System and policy makers in 2020. Given the Federal Reserve System's unprecedented direct purchase of financial assets, the financial sector exhibits a milder increase than the industrial sector.
- While the overall scenario-conditioned PD level for the Industrials subportfolio is much higher than the Financials level (3.0% vs. 1.5% respectively, during the economic growth period following the dot-com bust), variability is far higher for the Financials subportfolio.
- Compared to the Industrials subportfolio, the Financials subportfolio shows a significantly larger sensitivity to the stress that occurred during the aftermath of the financial crisis. We can explain higher scenario-based PD levels observed outside of economic downturns via the weighted-average EDF value for the Industrials subportfolio, which is approximately two times the weighted-average EDF value for the Financials subportfolio. In the case of the higher Financials variability, one of the primary drivers is the higher weighted average R-squared (40.53% for Financials vs. 29.84% for Industrials), which leads

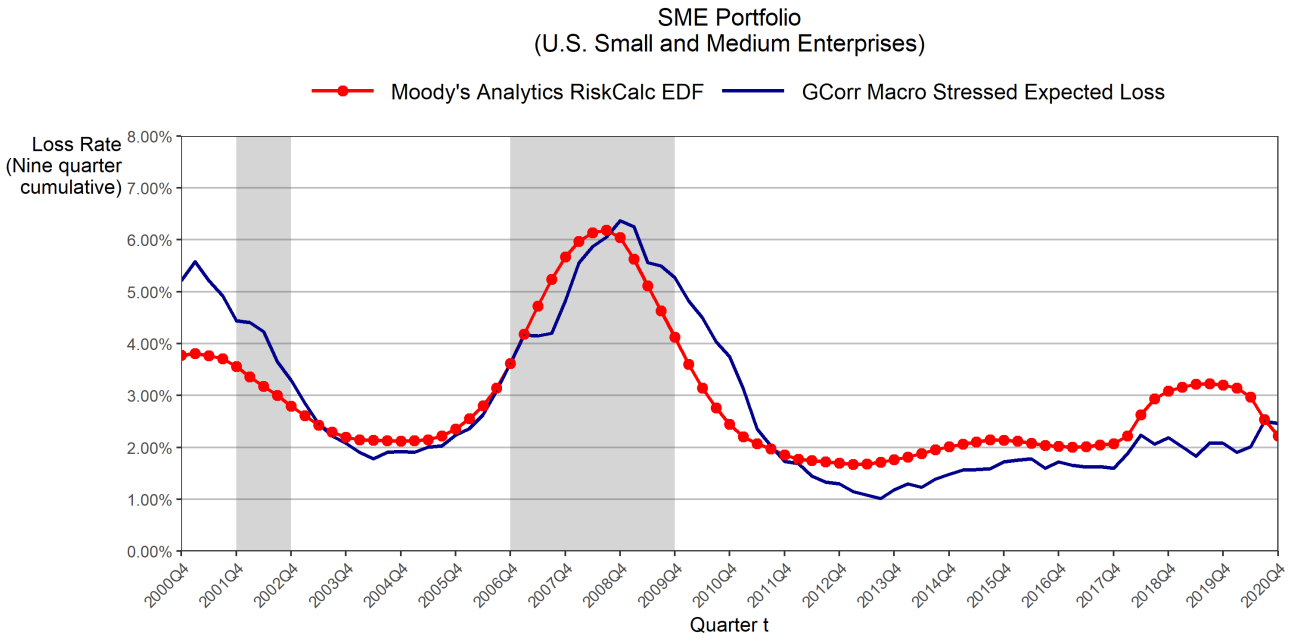
to higher sensitivity to macroeconomic shocks.

Benchmarking for U.S. SME Portfolio

In the next set of validation exercises, we study the U.S. SME portfolio, and we compare the expected losses produced by the GCorr Macro model to various benchmarks.

Moody's has developed a methodology for stressing PD values based on the RiskCalc modeling framework for private firm PD values[6]. We refer to this model as the RiskCalc EDF and we use it as a benchmark for the GCorr Macro model. The portfolio setup of both models remains the same. In Figure 9, we plot the time series of nine-quarter cumulative losses for the U.S. SME portfolio projected by the RiskCalc model and the GCorr Macro model. Note, losses are based on scenario-conditioned PD projections only; we assume LGD to be constant. An important observation is that the two models provide comparable levels of nine-quarter projected losses, as Figure 9 shows. Time series patterns of losses from the two models are similar. The RiskCalc EDF model and the GCorr Macro model respond similarly to different economic episodes, such as projecting higher losses for the period of financial crisis than for the recession during the early 2000s and Pandemic.

Figure 9 Cumulative, nine-quarter scenario-based losses for the SME portfolio: the GCorr Macro model, Moody's Stressed PD Model, and Unconditional Expected Loss with fixed LGD of 50%.



To summarize these validation exercises, first, we show how losses produced by the GCorr Macro model vary across economic episodes during the period 2001 through 2023. Second, we compare loss levels from the GCorr Macro model to various benchmarks, with an emphasis on the financial crisis period. Validation exercises demonstrate that the GCorr Macro model can differentiate economic episodes according to their severity. The cumulative losses it projects over nine quarters for pre-crisis portfolios under the financial crisis scenario are broadly in-line with the benchmarks. These conclusions are relevant for CCAR-style analyses, as financial institutions stress test their portfolios with the current risk parameters (in other words, parameters from 2022 through 2023), assuming a severely-adverse economic scenario created by the Federal Reserve, which is similar to the financial crisis episode.

Another GCorr Macro model feature underscored by the validation exercises is the model's ability to handle both large public firm portfolios and SME portfolios. Under the GCorr framework, we are able to account

[6]See the paper Chen et al. (2014).

for the different credit risk dynamics SME and large public firms are exposed to and in turn, their correlation with macroeconomic variables.

3.2 INTERNATIONAL CORPORATE PORTFOLIOS

In this section we present how the GCorr Macro model multi-period stress testing methodology performs for portfolios of large corporate firms in major countries including France, UK, Canada, Japan, and Brazil. Table 3 summarizes the stylized credit portfolios of the aforementioned regions.

In Table 3, for all portfolios:

- Types of counterparties: Large-listed corporates in a given region (firms constituting 99% of total liabilities of firms in the given region)
- Exposure pooling: By industries within a given region
- Pool weights: Proportional to the large firm counts by GCorr industries in GCorr 2023 Corporate
- Source of the R-squared values: GCorr 2023 Corporate, large firm average R-squared values by industry (applies to all portfolios)
- Definition of PD: Unconditional (input) PDs are time-varying, average pool-level EDF values from CreditEdge
- Loss Given Default for Projections: 100%

Table 3 Stylized Portfolios Used for Validation

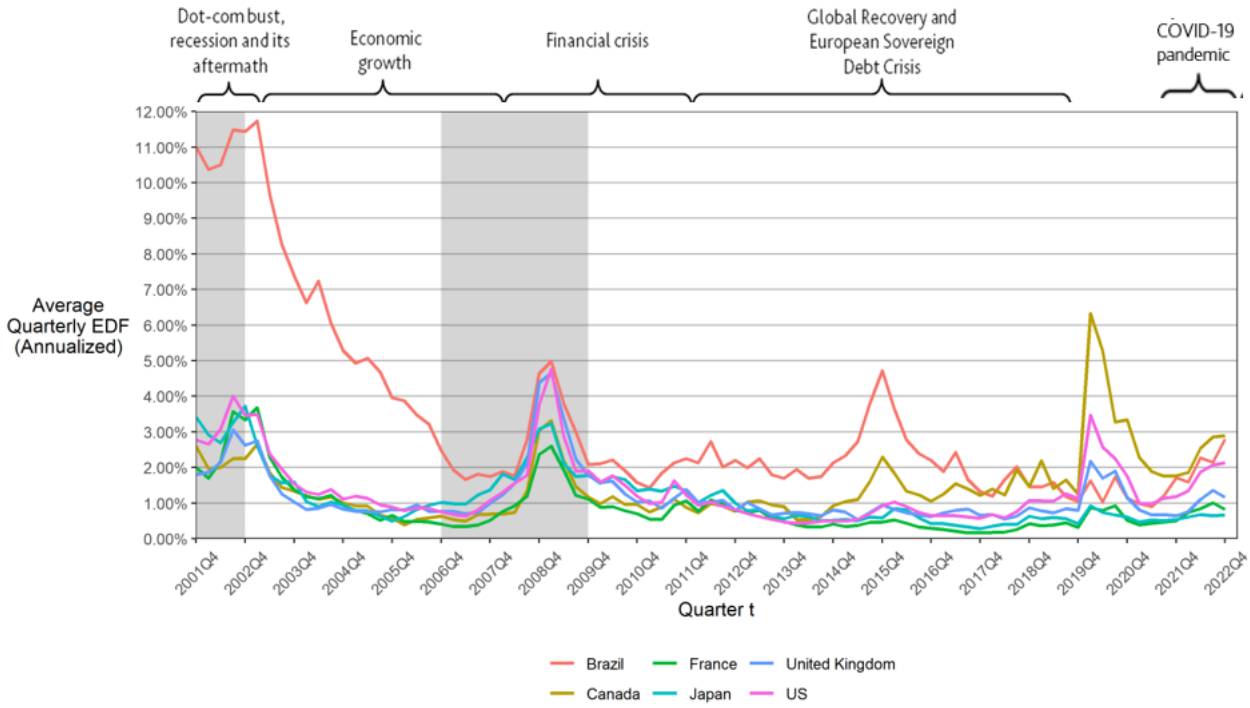
Portfolio	France Portfolio	UK Portfolio	Canada Portfolio	Japan Portfolio	Brazil Portfolio
Exposure Pooling	52 pools of loans to France corporates Loans	58 pools of loans to UK corporates Loans	52 pools of loans to Canada corporates Loans	60 pools of loans to Japan corporates Loans	46 pools of loans to Brazil corporates Loans
Weighted R-squared	28.07%	28.29%	28.71%	29.44%	31.45%

Comparing these portfolios with each other and the large corporate portfolio for the U.S., we see that the U.S. portfolio bears the greatest systematic risk (largest average R-squared), followed by those for Brazil and Japan. In terms of the default risk measured by the portfolio weighted average probability of default at the start of 2007, we find that the lowest risk by far is seen for the France portfolio. On the other hand, Japan, Brazil, and the U.S. exhibit high PD values compared to the other regional portfolios.

Figure 10 shows the average CreditEdge™ benchmark EDF value for large public firms for the U.S., France, the UK, Canada, Brazil, and Japan. Like the U.S., we again see large increases during the dot-com bust and the financial crisis. From 2010 to 2019, no other economic downturns lead to notable EDF value increases in the UK and in Japan. For example, the impact of the 2011 Japanese earthquake led to only smaller increases in average EDF levels. However, we do see a notable increase in the EDF time series starting from 2014 in the Canadian portfolio due to falling oil prices. During 2020, the immediate economic impact of the Great Lockdown due to the pandemic was mostly observed in the U.S and Canada. In contrast to the sharp increase in North America, a much milder reaction to the pandemic is observed in Brazil, which reflects the impact of different public policies.

Comparing the different country/regions during the dot-com bust and the financial crisis, we see that during the former, Japan and France have higher EDF levels. Overall, the EDF time series behavior from the U.S., UK, and Canada is affected during both crisis periods. The EDF value spikes for the other international regions are higher throughout. Japan and France EDF levels show the lowest increase during the financial crisis, with comparable levels during the height of the dot-com bust.

Figure 10 Average CreditEdge benchmark EDF for large France, UK, Canada, Japan, Brazil, and U.S. public firms.

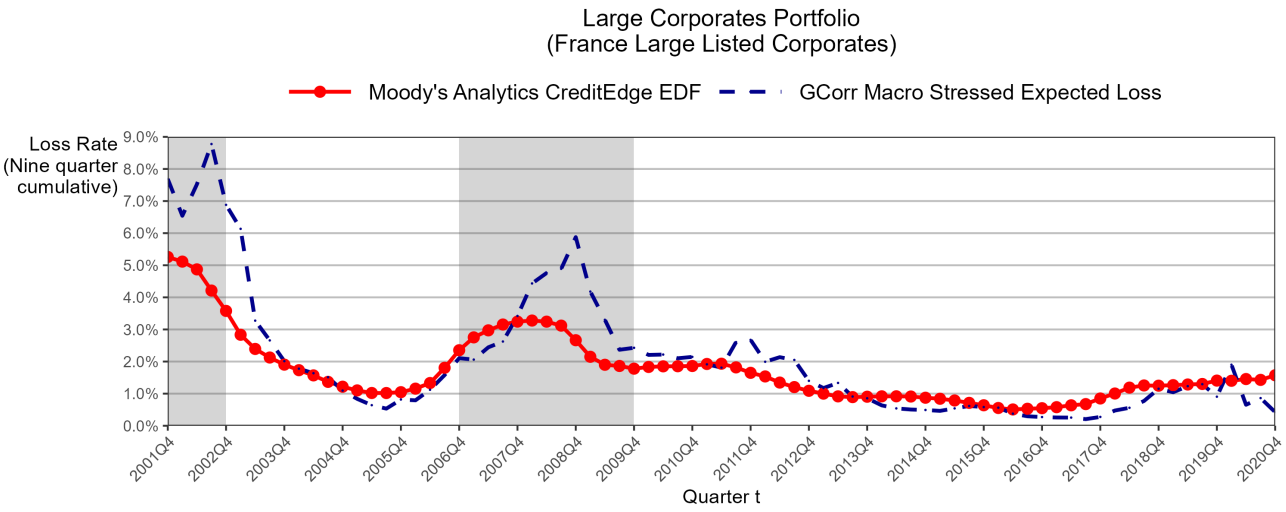


France

In the following analysis for France, we focus on the selected regional top model performance. The selected France model consists of France Equity, France GDP, and Eurozone Unemployment. Compared to the U.S., the adjusted pseudo R-squared of the model is lower (24% vs. 45%).

In Figure 11, we compare the backtesting results of the scenario-conditioned PD from the top France GCorr Macro model to the benchmark CreditEdge EDF measure.

Figure 11 Backtesting GCorr Macro model scenario-based loss for a France large corporates portfolio with time-varying PD term structure and LGD=100%.

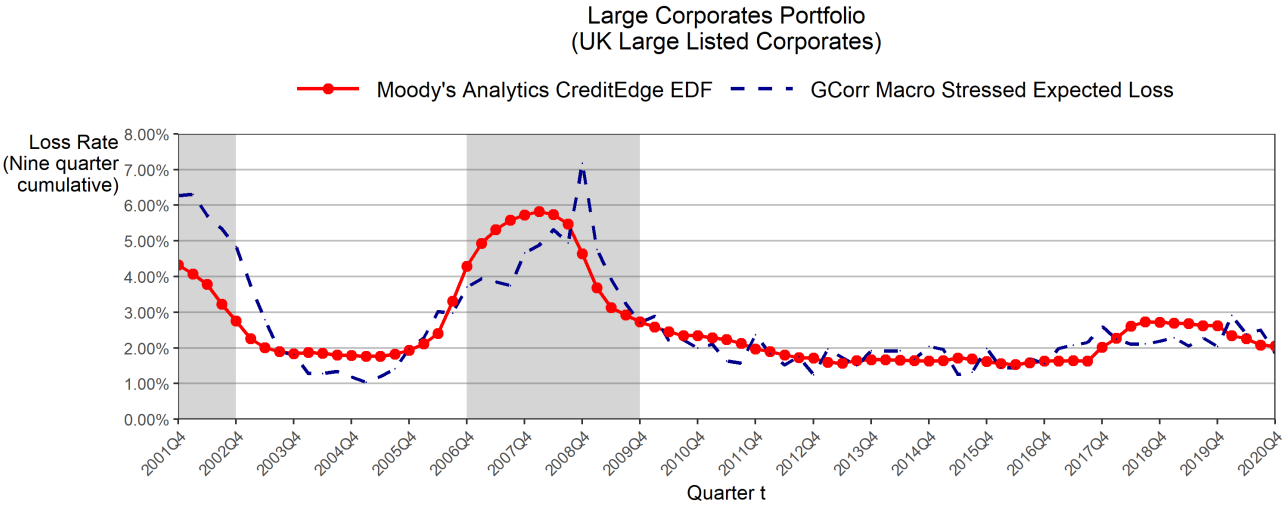


Looking at the pattern of the GCorr Macro model scenario-based PD values and comparing it to the EDF benchmark, we see that both time series are in general aligned during the time period studied. The scenario-based PD time series shows a higher increase and stronger recovery compared to EDF time series during the Global Financial Crisis (GFC), mainly driven by France GDP dynamics. It then exhibits a minor increase during the Eurozone crisis, followed by a recovery period, and finally increasing again during the pandemic, as can be seen in the nine-quarter cumulative periods starting in 2018Q4.

United Kingdom

The top model selected for United Kingdom[7] consists of UK GDP and UK Equity. The pseudo R-squared of the model is lower than that of the U.S. model (31% for UK vs. 45% for the U.S.).

Figure 12 Backtesting the GCorr Macro model scenario-based loss for a UK large corporates portfolio with time-varying PD term structure and LGD=100%.



In Figure 12, we see that the scenario-conditioned PDs line up very well with the realized EDFs. The scenario-conditioned PDs rise during the financial crisis to a level that is similar to the realized EDFs, and then subsequently subside to lower levels during the economic recovery post-2009. Dynamics are also aligned in the recent periods.

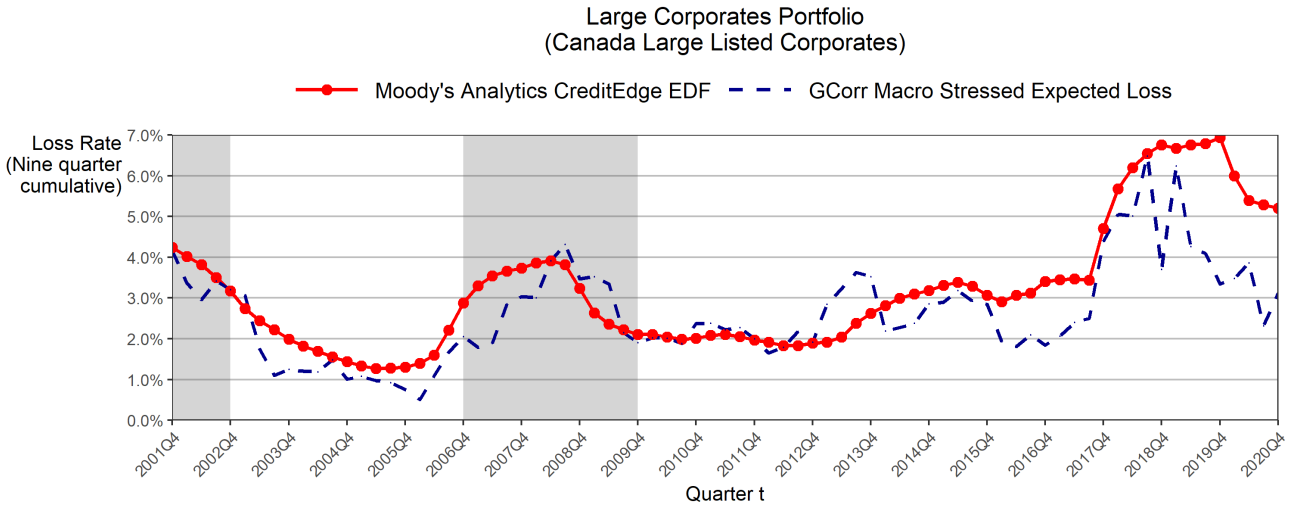
Canada

The Canadian model[8] consists of Canada Equity, Oil Price, and Canada Unemployment. The adjusted pseudo R-squared for the top model is 25%.

[7]UK Credit Spread is not supported by Economy.com, Moody's in-house data provider. As a result, the macroeconomic variable is not considered during the variable selection process for United Kingdom.

[8]Canada Credit Spread is not supported by Economy.com, Moody's in-house data provider. As a result, the macroeconomic variable is not considered during the variable selection process for Canada.

Figure 13 Backtesting the GCorr Macro model scenario-based loss for a Canadian large corporates portfolio with time-varying PD term structure and LGD = 100%.

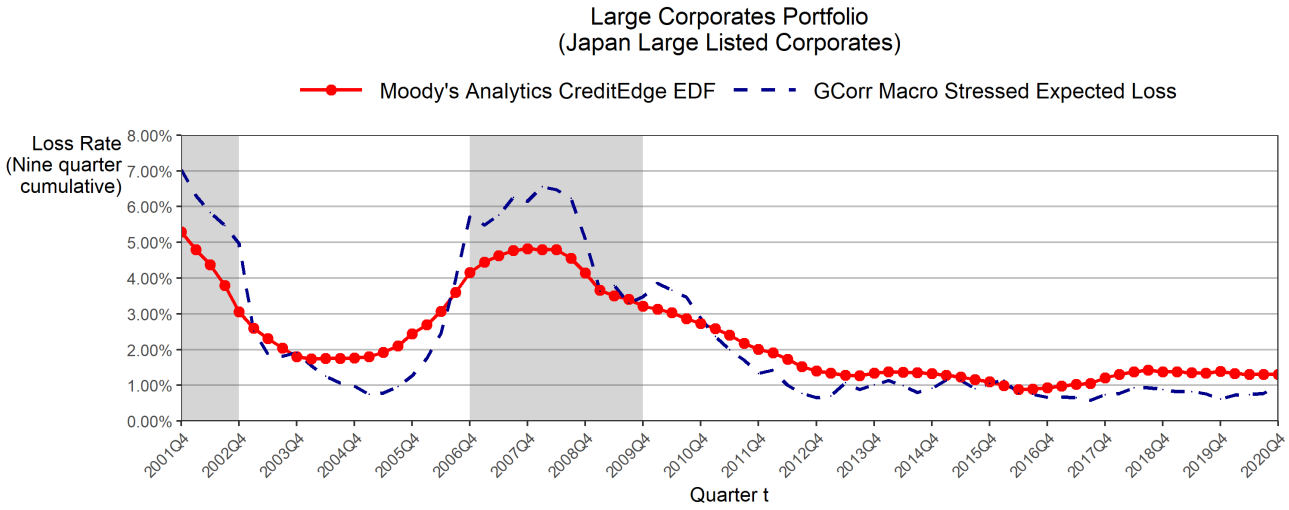


In Figure 13, the scenario-based PD projections for Canadian firms increased during the financial crisis, similar to the level of the EDFs. In recent periods, oil price has been declining. Since Canada is a large oil exporter, the oil drop is also reflected in the increase in the EDF and scenario-based PD projections.

Japan

The top model for Japan consists of Japan Equity and Japan GDP. The adjusted pseudo R-squared for the top model is 34%.

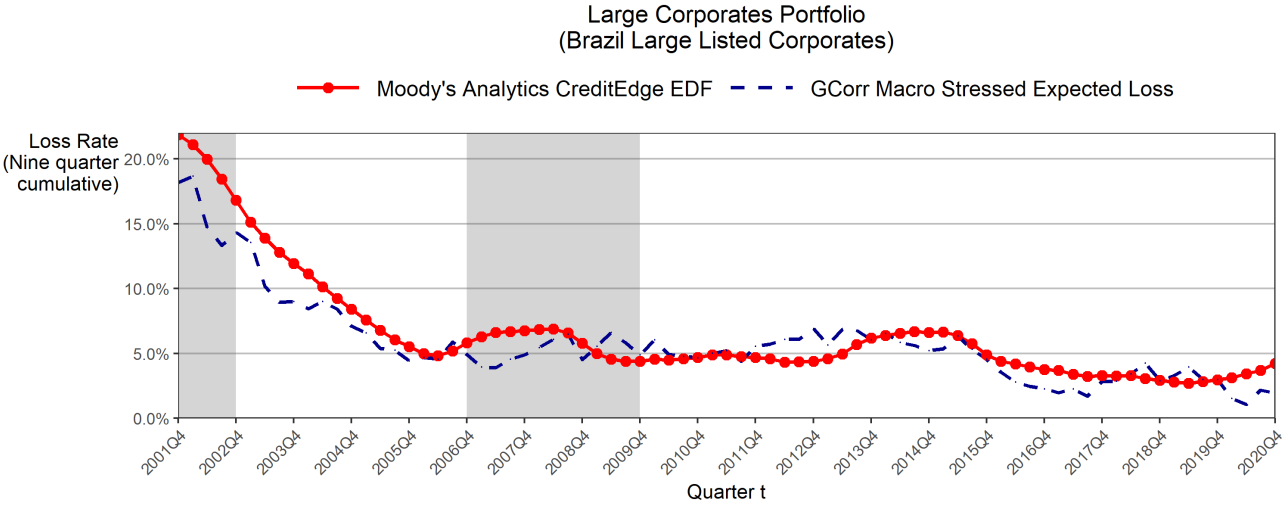
Figure 14 Backtesting the GCorr Macro model scenario-based loss for a Japanese large corporates portfolio with time-varying PD term structure and LGD=100%.



In Figure 14, we see that the scenario-conditioned PDs line up very well with the realized EDFs. The scenario-conditioned PDs rise during the financial crisis with GCorr stressed PD being slightly higher.

Brazil

Figure 15 Backtesting the GCorr Macro model scenario-based loss for a Brazil large corporates portfolio with time-varying PD term structure and LGD=100%.



The top model for Brazil consists of Brazil Equity, Brazil GDP, and Oil Price. Its adjusted pseudo R-squared is 38%. In Figure 15, the PD pattern of Brazil is very different from other countries. The stressed PD peaked during the dot-com bubble, and then decreased. We did not see a stressed PD spike during the financial crisis period.

The figures in this section show the historical performance of the GCorr Macro model for various countries. At each quarter, we compare the projected PD against the realized EDF. Emerging countries usually have fewer firms and noisier historical macrovariable data, which makes the exercise more challenging. For stress testing, one can look at the performance during the crisis periods, which are indicated by the shaded regions. For impairment calculations that involve average losses over a range of scenarios, one can look at the performance over several periods.

All off-the-shelf validation has been performed on Moody's constructed portfolios of publicly-traded firms. We re-commend additional validation exercises performed using an institution's own historical experience. More details can be found in Huang, Levy, and Pospisil (2017).

3.3 U.S. COMMERCIAL REAL ESTATE PORTFOLIOS

In this section, we demonstrate how to use the GCorr Macro model for non-corporate asset classes, namely for portfolios of U.S. Commercial Real Estate (CRE) exposures.

We first focus on CRE exposures. Figure 16 summarizes correlations of U.S. GCorr CRE factors with U.S. macroeconomic variables: Real GDP, U.S. Equity Index, and CRE Price Index[9]. For each property type (Hotels, Industrial, Multi-Family, Office, and Retail), the correlations vary across 374 U.S. Metropolitan Statistical Areas (MSAs). All three variables have significantly positive correlations with the factors, which is consistent with economic intuition. For example, economic conditions measured by Real GDP growth and direction of the commercial real estate market measured by CRE Price Index return affect a CRE portfolio's performance.

[9]GCorr CRE contains 374 MSA factors and five property type factors (Hotels, Industrial, Multifamily Housing, Office, and Retail) to measure systematic risk for commercial real estate properties.

Figure 16 Distribution of correlations of U.S. GCorr CRE factors and select U.S. macroeconomic variables.

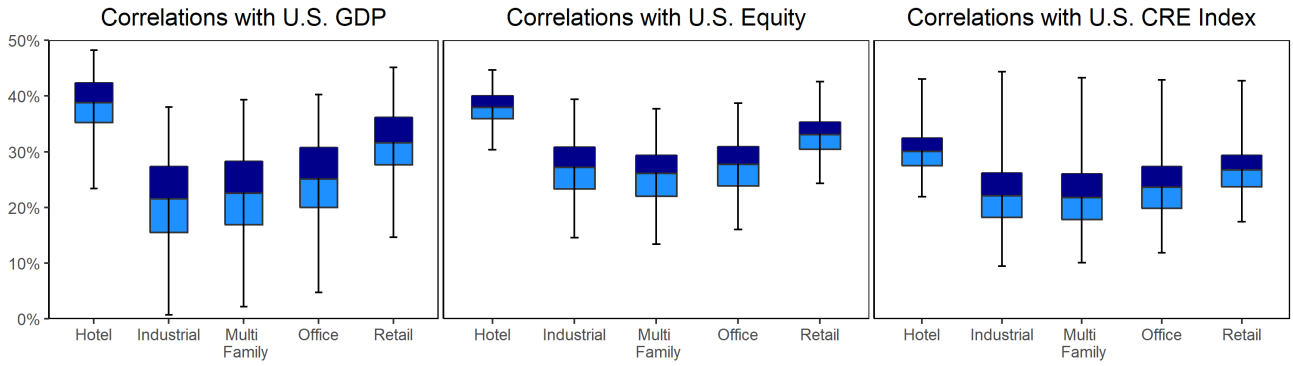


Table 4 summarizes the stylized credit portfolios we use to validate the GCorr Macro model for U.S. CRE. Unlike corporate portfolios, where point-in-time CreditEdge EDF measures for large corporates are well developed, setting the level of probability of default in a CRE portfolio over time can be quite challenging. For this exercise, we assume a constant PD through time to backtest model performance with a CRE portfolio. Pool-level probability of default is set to the pre-crisis default rate in 2008Q2, so we can understand how the model behaves during a financial crisis.

Table 4 Stylized Portfolios Used for Validation

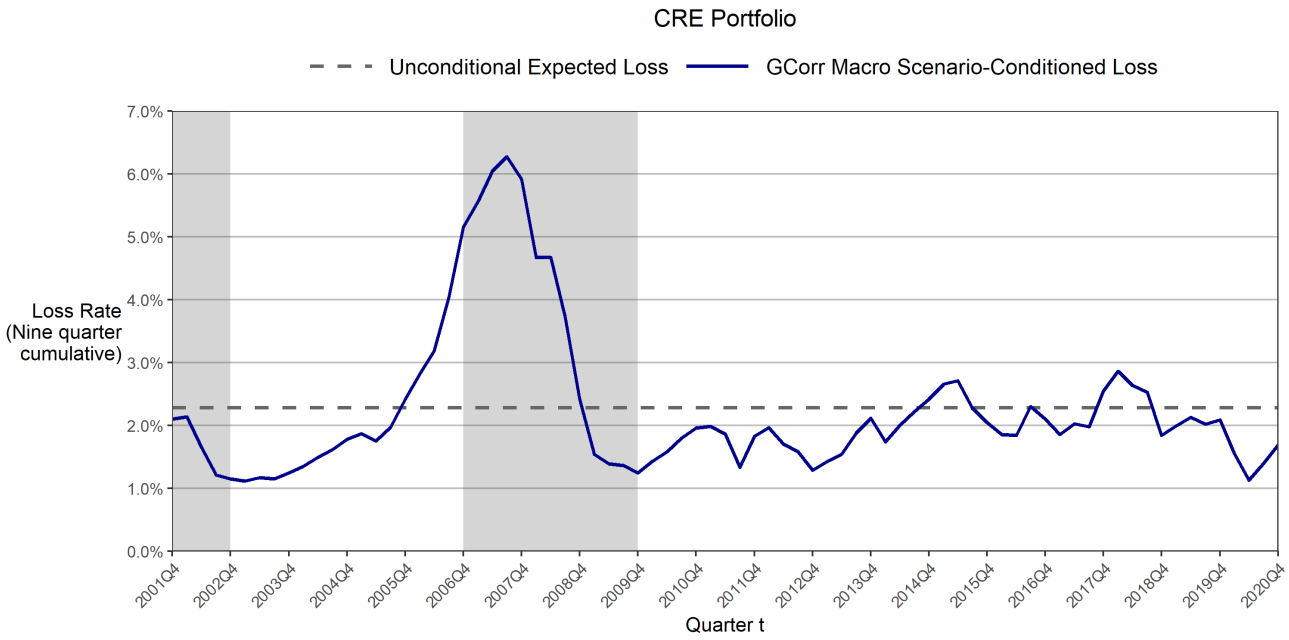
PORTFOLIO	CRE PORTFOLIO
Types of Counterparties	Commercial Real Estate Debtors
Exposure Pooling	1870 pools of loans Loans are pooled by 374 GCorr MSAs for each product
Pool Weights	Pool weights proportional to the CMBS initial balance by GCorr MSA in GCorr 2023 CRE
R-squared	Source: GCorr 2023 CRE Weighted average R-squared = 35.26%
Probability of Default	Fixed PD[a]: Weighted average PD = 1.07% (annualized)
Loss Given Default for Projections	LGD = 100%

[a] Empirical RMBS default rate.

Figure 17 shows that the level of GCorr Macro model stressed losses peaks at 6%, which roughly matches the peak of the historical default rate during the financial crisis (we do not display the historical default rate in this documentation due to data licensing restrictions[10]). But the peak of the GCorr Macro model losses occurred in 2007Q3, preceding the historical nine-quarter accumulated default rate, which peaked in early 2009. One explanation for the timing difference is that the GCorr Macro model exhibits the highest losses in the nine-quarter period of the most adverse economic shocks, while the realized loss by its nature is less fluctuating, and often lags behind economic indicators.

[10]The default rate is defined as 90 days to 120 days delinquency rate in the projection.

Figure 17 Backtesting of the GCorr Macro model scenario-based loss from the GCorr Macro model for a U.S. CRE portfolio with constant PD term structure and LGD=100%.



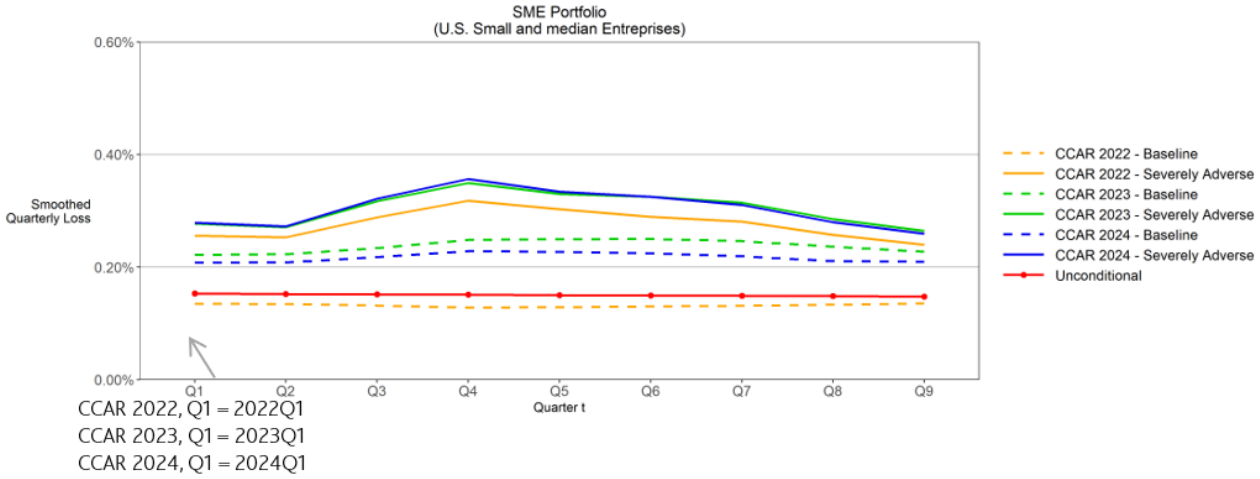
4. Projected CCAR Losses Based on the GCorr Macro Model

This section presents the results of several stress testing exercises. We use the portfolios introduced in the previous section to calculate the stressed expected losses with the GCorr Macro model under the CCAR scenarios. The questions of interest are how the projected losses compare across scenarios and across portfolios, as well as how they compare to the losses based on the historical scenarios described in section 3.

4.1 U.S. LARGE CORPORATE AND SME PORTFOLIOS

The Federal Reserve publishes two CCAR scenarios each year: Baseline and Severely Adverse[11]. We chose three macroeconomic variables for each portfolio type to project losses with the GCorr Macro model under the CCAR scenarios: * For the U.S. Large Corporates portfolio: U.S. Unemployment Rate, U.S. BBB Spread, and Dow Jones Total Stock Market Index * For the U.S. SME portfolio: U.S. Unemployment Rate, U.S. GDP, and Dow Jones Total Stock Market Index Figure 18 and Figure 19 show the quarterly smoothed stressed expected losses for the U.S. SME portfolio and the U.S. Large Corporates portfolio, respectively. We use the same portfolios as in section 3.

Figure 18 Quarterly smoothed losses projected by the GCorr Macro model for a U.S. SME portfolio under CCAR 2022, CCAR 2023, and CCAR 2024 scenarios.



[11] In additional to the two supervisory scenarios ("Baseline" and "Severely Adverse"), there are two exploratory scenarios that look at funding stress in CCAR 2024. The two exploratory scenarios are not shown in the documentation as the exploratory analysis will not impact large bank capital requirements.

Figure 19 Quarterly smoothed losses projected by the GCorr Macro model for a large corporates portfolio under CCAR 2022, CCAR 2023, and CCAR 2024 scenarios.

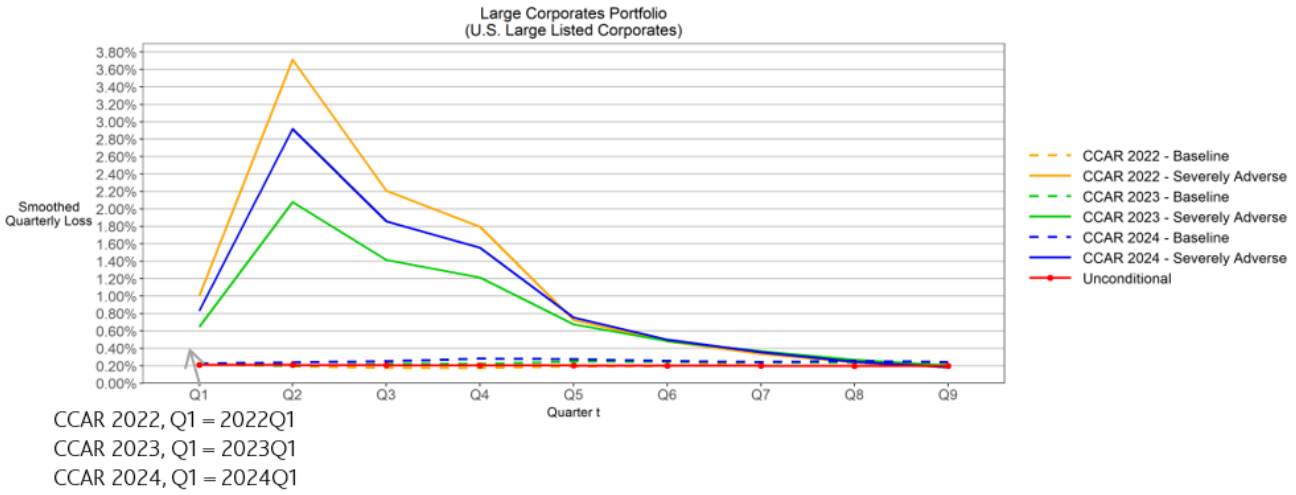


Figure 18 and Figure 19 exhibit similar patterns. The projected quarterly losses under the Severely Adverse scenarios are higher during the earlier quarters. We can attribute this finding to the assumed paths of the macroeconomic variables; the shocks occur early, which leads to large quarterly losses at the beginning, and the variables recover in later quarters, implying lower losses at the end.

Figure 20 presents the log changes for the three macroeconomic variables, under the CCAR 2022, CCAR 2023, and CCAR 2024 Severely Adverse scenarios.

Figure 20 Quarterly log changes of selected macroeconomic variables shocks under the CCAR 2022, CCAR2023, and CCAR 2024 Severely Adverse scenarios.



Table 5 describes the nine-quarter cumulative losses for both SME and Large Corporate portfolios under the different CCAR scenarios. In-line with economic intuition, the Severely Adverse scenarios are associated with the largest projected losses among the scenarios, with losses from CCAR 2022 being slightly higher than others for Large Corporate portfolio. As for the Baseline scenarios, all portfolios exhibit similar cumulative losses across CCAR scenarios.

Comparing the paths of macroeconomic variables in Figure 20, the Severely Adverse scenario in CCAR 2024 is the most severe due to the sharp increase in credit spread and the ramp-up of the U.S. stock market in the first quarter. An extreme increase of 50% in US unemployment rate implies a distinct difference in CCAR 2022's unemployment prediction.

Table 5 Nine-quarter Cumulative Expected Losses CCAR 2022, CCAR 2023, CCAR 2024 Scenarios for U.S. SME and Large Corporates Portfolios

SCENARIO	SME	LARGE CORPORATES
Unconditional	1.35%	1.83%
CCAR 2022 - Baseline	1.19%	1.87%
CCAR 2022 - Severely Adverse	2.48%	10.68%
CCAR 2023 - Baseline	2.13%	2.03%
CCAR 2023 - Severely Adverse	2.73%	7.36%
CCAR 2024 - Baseline	1.95%	2.28%
CCAR 2024 - Severely Adverse	2.74%	9.21%

Note that the CCAR Baseline scenario losses are not substantially lower than the unconditional losses in

Figure 18 and Figure 19. To explain this pattern, we examine the macroeconomic variable paths under the scenarios. Although the unemployment rate decreases, the stock market provides positive returns, and the BBB Spread decreases slightly.

Table 5 shows an interesting feature. The losses across scenarios are more dispersed for the U.S. Large Corporates portfolio than for the SME portfolio. As we point out in the context of the historical scenarios in section 3, we can attribute this effect to the substantially different asset R-squared values of the portfolios: 34.73% for the Large Corporates portfolio and 7.17% for the SME portfolio.

4.2 U.S. COMMERCIAL REAL ESTATE PORTFOLIOS

In this section we use the same set of variables (Real GDP, U.S. Equity, CRE index) and portfolios as in section 3, except that we set LGD equal to 40%. Figure 21 shows the quarterly-smoothed-stressed expected losses for the U.S. CRE portfolio.

The CCAR 2022 Severely-Adverse scenario leads to losses that are much higher than the losses obtained from the Severely-Adverse scenario in CCAR 2023 and CCAR 2024. This follows from the fact that both the equity and the CRE index returns in the CCAR 2022 Severely-Adverse scenario are more severe than those from other CCAR scenarios.

Figure 21 Quarterly smoothed losses projected by the GCorr Macro model for the U.S. CRE portfolio under CCAR 2022, CCAR 2023, and CCAR 2024 scenarios.

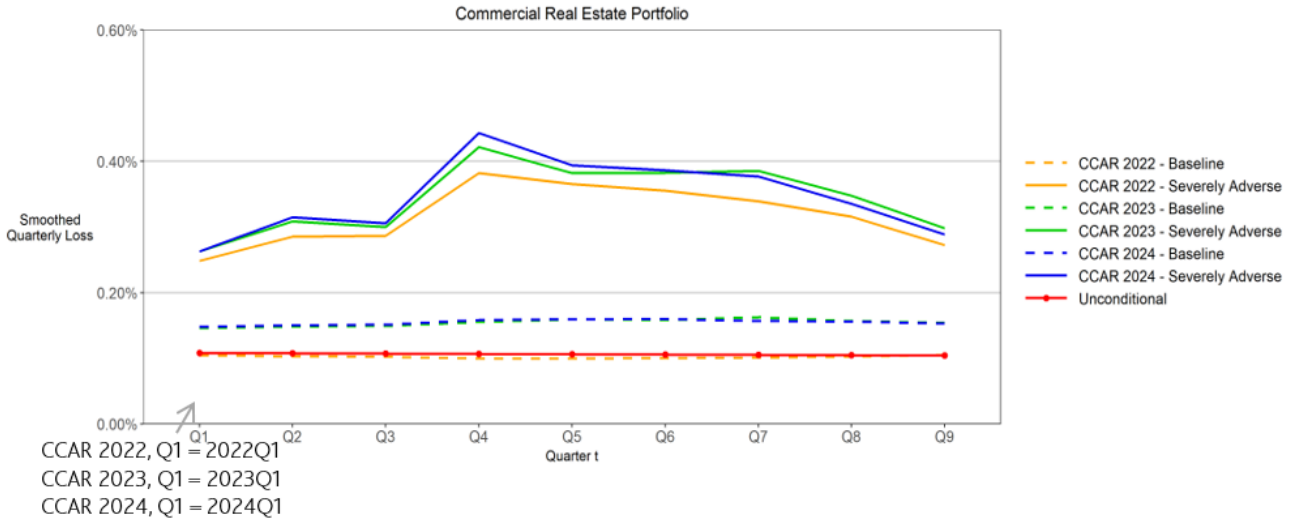


Table 6 Nine-quarter Cumulative Expected Losses for CCAR 2022, CCAR 2023, and CCAR 2024 Scenarios for U.S. CRE Portfolio

Scenario	CCAR 2022	CCAR 2023	CCAR 2024
Unconditional	0.96%	0.96%	0.96%
Baseline	0.92%	1.39%	1.39%
Severely Adverse	2.85%	3.09%	3.11%

5. Summary

Moody's GCorr Macro model links borrower credit quality to macroeconomic scenarios, enabling multi-period PD and LGD value projections under various macroeconomic scenarios. In this paper, we presented several validation exercises in which we used GCorr Macro model scenario conditioning to produce losses on Commercial & Industrial, SME, and CRE portfolios under various historical scenarios. As our results indicate, time series dynamics of these projected losses are in line with economic intuition, and the loss levels are appropriate when compared to various benchmarks.

The relationships in the GCorr Macro model are calibrated based on historical data from 1999 and are most suitable when applying macrovariable scenarios that are in line with historical experiences during this period. The GCorr 2023 Macro model offers several updates from the previous model year by extending the data to March 2023. With addition of 9 months of data being added in the empirical study, some overall pattern changes emerged:

- GCorr 2023 asset correlations are on average slightly lower than GCorr 2022 asset correlations and have receded to pre-pandemic levels.
- Milder relationships between macro-variables and credit factors, especially in gross domestic product (GDP) and unemployment types of macroeconomic variables, have been observed. As a result, the overall correlation levels have receded to pre-pandemic levels.

A. Variable Selection Results for Corporate Exposures

To run scenario-based calculations, we first choose the set of macrovariables to be included in the analysis. Moody's has performed a variable selection procedure for corporate exposures in each individual country. We describe a variable selection procedure that relies upon standard regression model techniques in the GCorr Macro model context.

The set of selected macroeconomic variables should meet several criteria:

- The set should statistically explain a sufficient portion of variation in the systematic credit risk factors.
- The model must be parsimonious, in the sense that it must achieve high explanatory power with as few variables as possible. This allows the set to avoid multicollinearity and reduce noise in parameter estimates.
- There should be an economic narrative explaining why the selected variables are relevant for the given portfolio. This includes ensuring that the direction and strength of the relationship between each variable in the model and portfolio losses is in-line with economic intuition.

For each individual country, we first select the subset of candidate macrovariables to include in the selection procedure. We first include each country's equity, unemployment rate, and GDP. For European countries, we include Eurozone GDP and Eurozone Unemployment Rate. For oil-producing countries, we also include Oil Price.

We then calculate the different models, combining candidate macrovariables. For each combination, we determine the coefficients of the instruments' systematic credit risk factors to the macroeconomic factors included in the model, the corresponding t-statistics, and the adjusted multivariate correlation. Of all the candidate models, we exclude those that fail at least one of the following two tests:

1. At least one estimated coefficient in the model is insignificant according to its t-statistic. This restricts the number of variables in the model to only the ones that contribute to explaining variation in the credit risk factors, and, thus, keeps the model parsimonious.
2. At least one coefficient has an unintuitive sign. This eliminates models with economically-unintuitive relationships. We rank the models that pass the two criteria according to their explanatory power measured by their adjusted multivariate correlation. The adjusted multivariate correlation captures the trade-off between how well the model fits and the number of parameters estimated.

The following table shows the statistics of the default country model for the U.S.and other major countries.

Table 7 Sample Variable Selection Results for GCorr Macro 2023

Country	Variable1	Variable2	Variable3	Variable4	Coef1 (T-stat)	Coef2 (T-stat)	Coef3 (T-stat)	Coef4 (T-stat)	Adjusted MultiCorr
Australia	Australia GDP	Australia Equity			0.217 (2.35)	0.428 (4.625)			0.533
Brazil	Brazil GDP	Oil Price	Brazil Equity		0.162 (1.859)	0.194 (2.331)	0.462 (5.163)		0.613
Canada	Canada Equity	Oil Price	Canada Unemployment		0.339 (3.658)	0.185 (2.076)	-0.221 (-2.41)		0.502
China	China Equity	China GDP			0.428 (4.632)	0.218 (2.362)			0.533
France	France Equity	France GDP	Eurozone Unemployment		0.323 (3.398)	0.18 (1.71)	-0.164 (-1.569)		0.495
Germany	Germany Equity	Germany GDP	Eurozone Unemployment		0.493 (5.436)	0.117 (1.195)	-0.118 (-1.212)		0.596

Hong Kong	Hong Kong Equity			0.585 (7.036)			0.579
Japan	Japan GDP	Japan Equity		0.122 (1.381)	0.541 (6.112)		0.584
Mexico	Mexico GDP	Mexico Equity		0.17 (1.784)	0.381 (3.993)		0.442
Netherlands	Netherlands Equity	Netherlands GDP	Eurozone Unem-ploy-ment	0.403 (4.31)	0.156 (1.526)	-0.136 (-1.339)	0.538
South Africa	South Africa GDP	South Africa Equity		0.246 (2.588)	0.349 (3.672)		0.472
Spain	Spain Equity	Spain GDP	Spain Unem-ploy-ment	0.452 (4.84)	0.053 (0.529)	-0.129 (-1.292)	0.507
Sweden	Sweden Equity	Eurozone Unem-ploy-ment		0.534 (6.043)	-0.139 (-1.579)		0.588
United Arab Emirates	Oil Price	United Arab Emirates Equity		0.456 (5.443)	0.317 (3.784)		0.622
United Kingdom	UK GDP	UK Equity		0.152 (1.673)	0.494 (5.451)		0.553
United States	US Unemployment	US Equity	US BBB Spread	-0.217 (-2.69)	0.313 (3.324)	-0.353 (-3.97)	0.671

B. List of Macroeconomic Variables for Sovereign Countries

Unlike GCorr Corporate's statistical variable selection process, we determine the relevant macroeconomic variables for each sovereign based on the interaction of sovereign credit risk with various macroeconomic variables, the interaction between macroeconomic variables, and the dependencies of sovereign financing. There are three general types of macroeconomic variables considered for sovereign countries. We first include each country's equity and unemployment rate to represent its general economic condition. For oil and commodities exporters, we include oil price and commodity price as the second macroeconomic variable. For countries with a large proportion of foreign-currency denominated debt, we include foreign exchange rate (USD based) as the third macroeconomic variable. While Eurozone countries are exposed to common shocks through their economic integration, for projection purposes, the model assumes that Eurozone shocks translate into macroeconomic variable shocks for individual countries as well. The following table shows the set of macroeconomic variables we consider in modeling for each individual sovereign country.

Table 8 Sample Sovereign MEV sets for GCorr Macro 2023

Country	Variable1	Variable2	Variable3	Variable4
Australia	Australia Unemployment	Australia Equity	Commodit Index	
Canada	Canada Equity	Canada Unemployment	Commodity Index	
China	China Equity	China Unemployment		
Finland	Finland Equity	Finland Unemployment		
France	France Unemployment	France Equity		
Germany	Germany Unemployment	Germany Equity		
Italy	Italy Equity	Italy Unemployment		
Japan	Japan Equity	Japan Unemployment		
Korea, South	Korea Equity	Korea Unemployment		
Luxembourg	Luxembourg Equity	Luxembourg Unemployment		
Netherlanc	Netherlanc Unemployment	Netherlanc Equity		

Singapore	Singapore Equity	Singapore Unemployment		
South Africa	South Africa Equity	South Africa Unemployment	Commodit Index	South Africa FX (USD/ZAR)
Spain	Spain Unemployment	Spain Equity		
United Kingdom	UK Equity	UK Unemployment		
United States	US Unemployment	US Equity		

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